

**Department of Computer Science & Engineering-Allied Branches**

**Operating Systems-24BECSE303**

**Module 5-Disk Scheduling Algorithm Problems**

**Problem1:**

Queue has the following requests with cylinder numbers as follows:

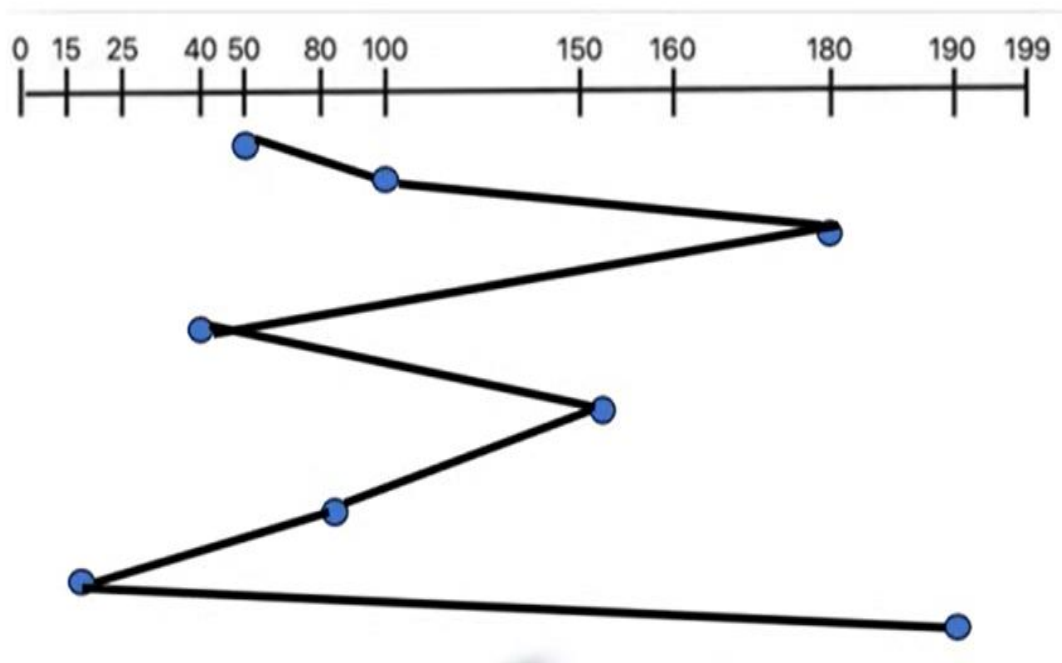
100, 180, 40, 150, 80, 15, 190

Assume the head is initially at cylinder 50. Calculate total head movement. Apply FCFS, SSTF, SCAN, C-SCAN, LOOK and C-LOOK Disk Scheduling Algorithm for the given problem.

**Solution:**

**Type1: FCFS Algorithm**

The head moves in the given order in the queue i.e., **50→100→180→40→150→80→15→190**



✓ **Total overhead movement (Seek Time) =**

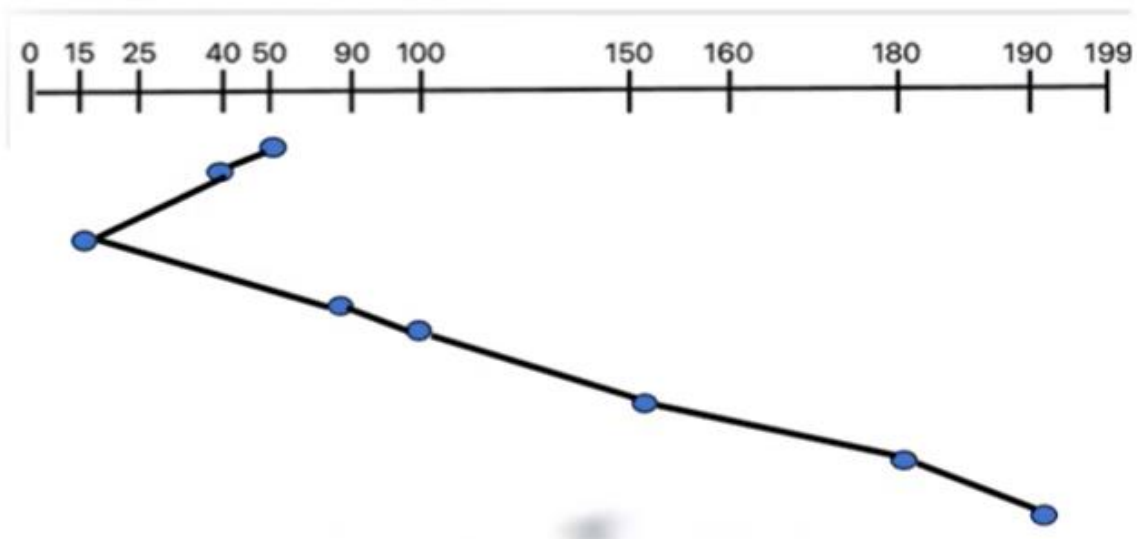
$$(100-50)+(180-100)+(180-40)+(150-40)+(150-80)+(80-15)+(190-15)$$

**Ans: =690**

### **Type2: SSTF Algorithm**

-This selects the request with minimum seek time from the current head position.  
SSTF chooses the pending request closest to the current head position.

-Assume the head is initially at cylinder 50. The next closest cylinder to 50 is 40, and then the next nearest one is 15, then 80, 100, 150, 180, 190.



✓ **Total overhead movement (Seek Time) =**

$$(50-40)+(40-15)+(90-15)+(100-90)+(150-100)+(180-150)+(190-180)$$

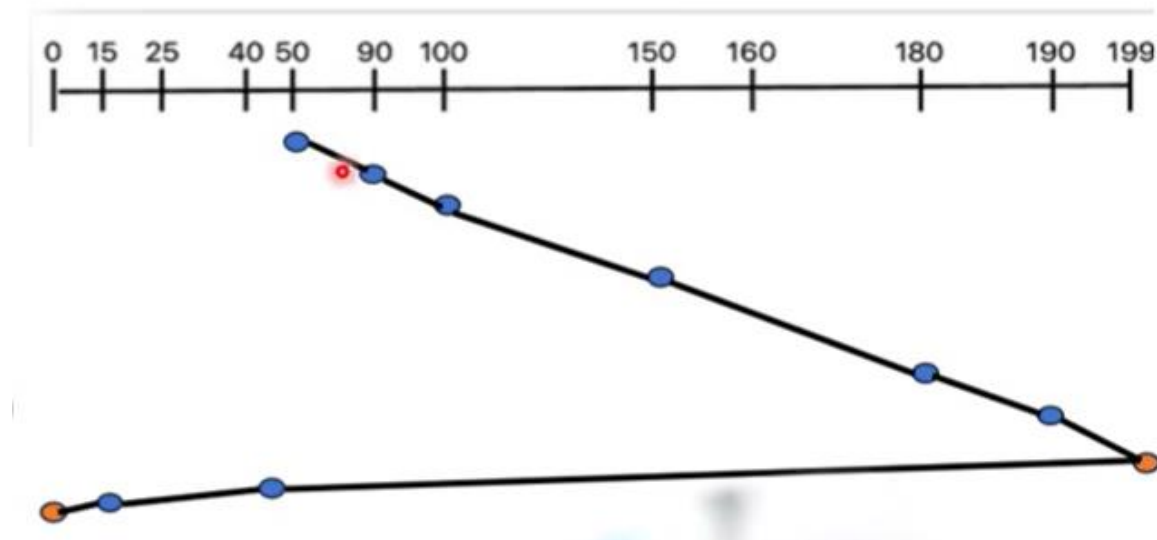
**= 210.**

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### **Type3: SCAN Algorithm**

-If the disk head is initially at 50 and if the head is moving towards forward direction, then it services 80, 100, 150, 180, 190 till 199 data included.

-At cylinder 199 the arm will reverse and will move towards the other end of the disk servicing 40 and then 15. The SCAN is also called as elevator algorithm.



✓ **Total overhead movement (Seek Time) =**

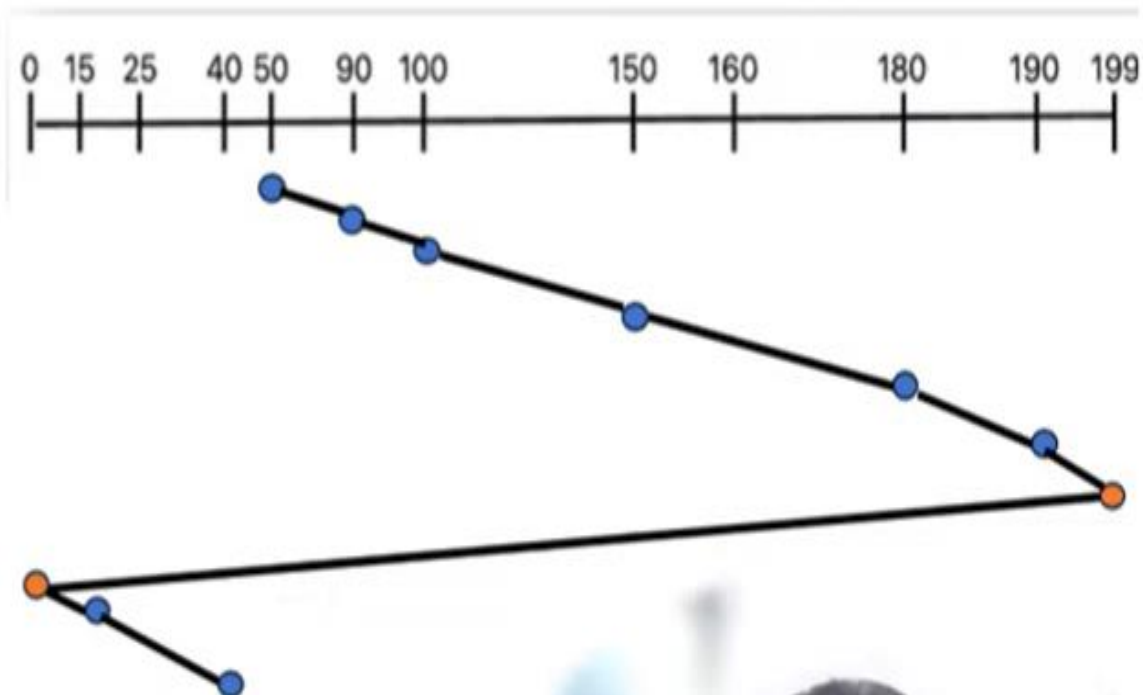
$$(90-50)+(100-90)+(150-100)+(180-150)+(190-180)+(199-190)+(199-40)+(40-15)$$

**Ans: =333**

#### **Type4: C-SCAN Algorithm**

-If the disk head is initially at 50 and if the head is moving towards the outer track, it services 80, 100, 150, 180, 190 and 199[reaches till end of the track].

-At cylinder 199 the arm will reverse immediately returns to the beginning of the disk without servicing any requests on the return trip (see chart below) and starts servicing again once reaches the beginning [including starting track zero] which serves 15 then 40.



✓ Total overhead movement (Seek Time) =

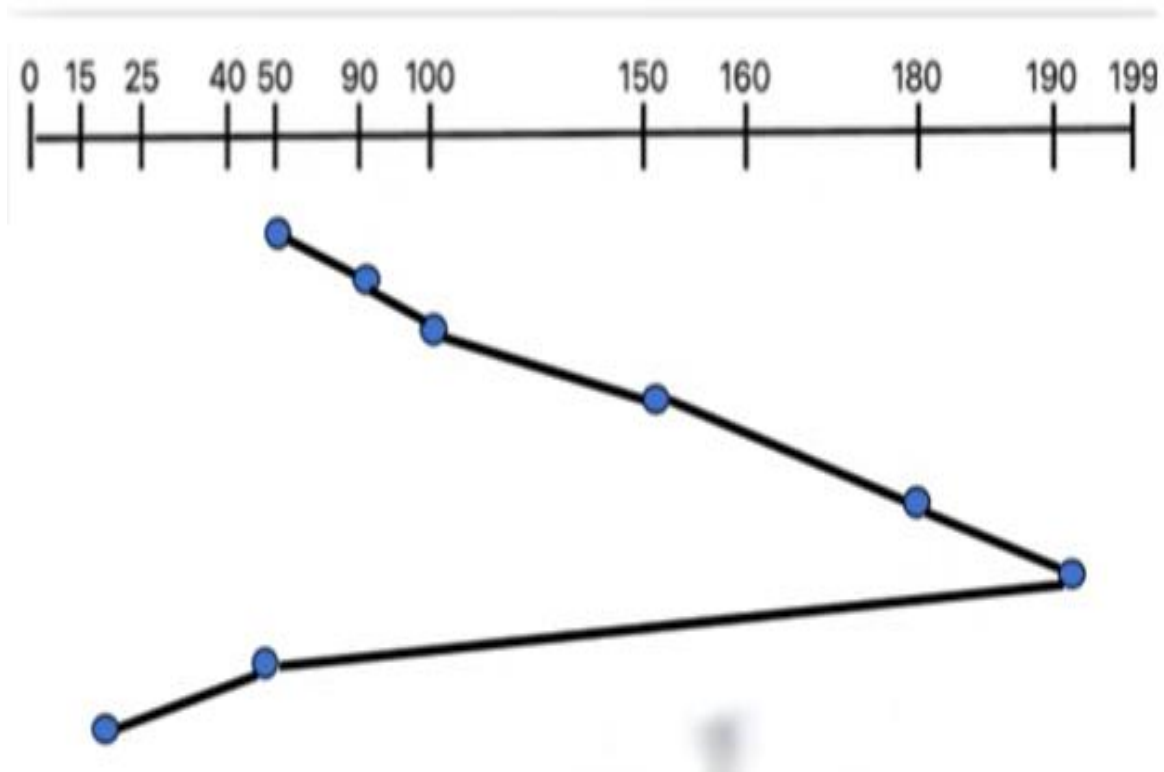
$$(90-50)+(100-90)+(150-100)+(180-150)+(190-180)+(199-190)+(199-0)+(15-0)+(40-15)$$

$$= 388$$

### Type5: LOOK Algorithm

-If the disk head is initially at 50 and if the head is moving towards the outer track, it services 80, 100, 150, 180, 190[Here it does not reach 199 like SCAN].

-At the final request 190, the arm will reverse and will move towards the first request 40 and then serves 15[does not reach end track].



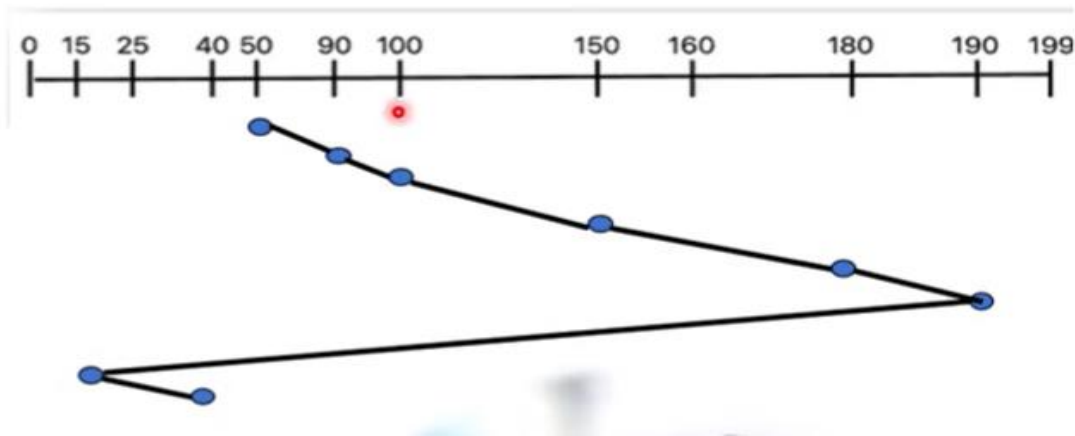
✓ **Total overhead movement (Seek Time) =**  
 $(190-50)+(190-15)$   
**=315**

### **Type6: C-LOOK Algorithm**

-Here the arm goes only as far as the final request in each direction. Then it reverses, without going all the way to the end of the disk.

-If the disk head is initially at 50 and if the head is moving towards the outer track, it services 80, 100, 150, 180, 190 without reaching end track 199.

-At the final request 190, the arm will reverse and will move towards the beginning of the first request 15 without starting from track zero and then serves 40.



✓ **Total overhead movement (Seek Time) =**  
 $(190-50)+(190-15)+(40-15)$

**Ans: =340**

**Problem2:** Queue has the following requests with cylinder numbers as follows:

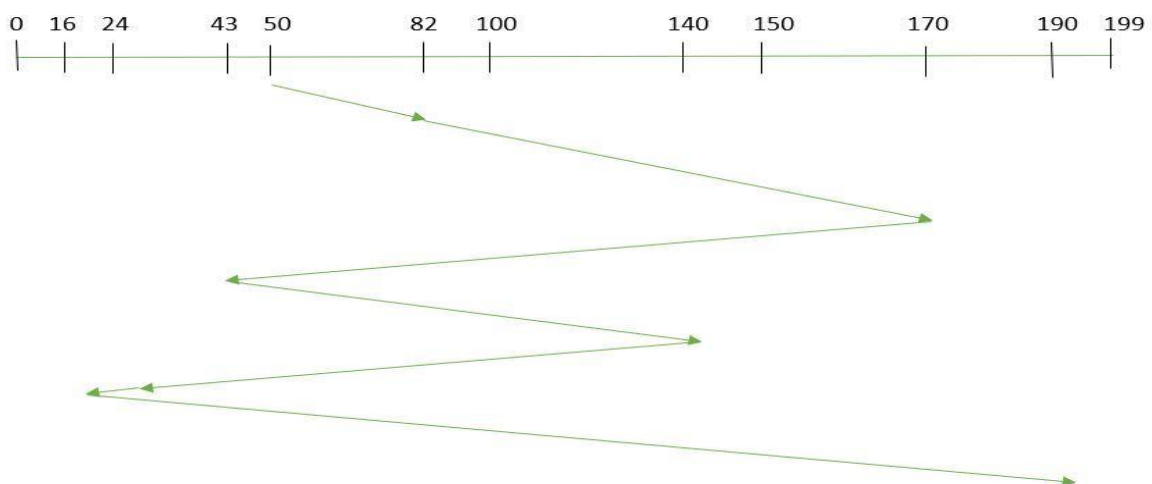
82,170,43,140,24,16,190

-Assume the head is initially at cylinder 50. Calculate total head movement. Apply FCFS, SSTF, SCAN, C-SCAN, LOOK and C-LOOK Disk Scheduling Algorithm for the given problem.

**Solution:**

### **Type1: FCFS Algorithm**

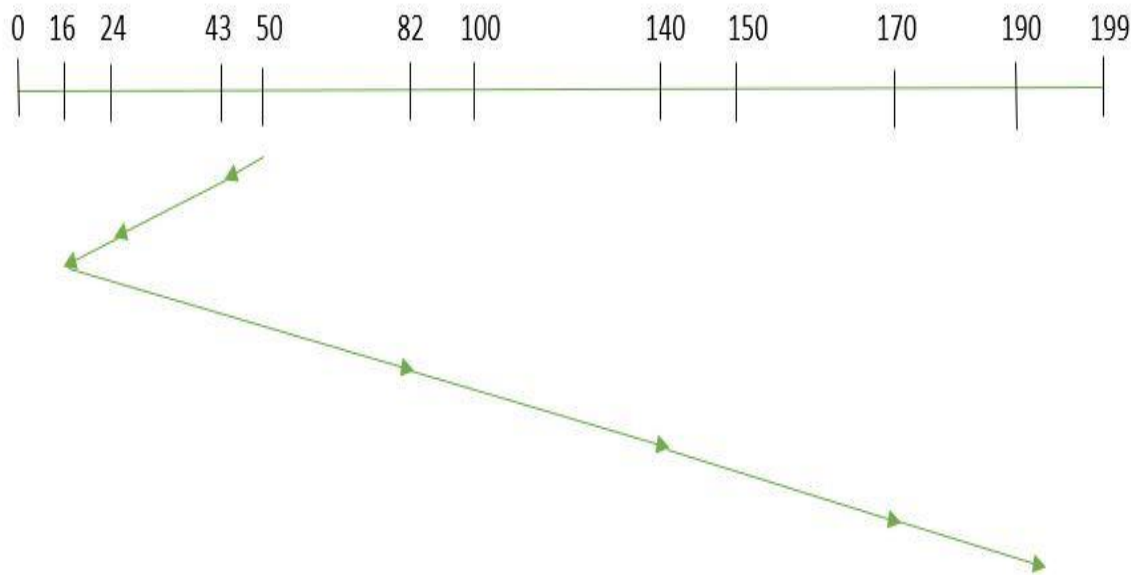
The head moves in the given order in the queue i.e., **50→82→170→43→140→24→16→190**



**Total OverHead Movement(Seek Time)=  $(82-50)+(170-82)+(170-43)+(140-43)+(140-24)+(24-16)+(190-16) = 642$**

### **Type2: SSTF Algorithm**

Assume the head is initially at cylinder 50. The next closest cylinder to 50 is 43, and then the next nearest one is 24, then 16, 82, 140, 170, 190.

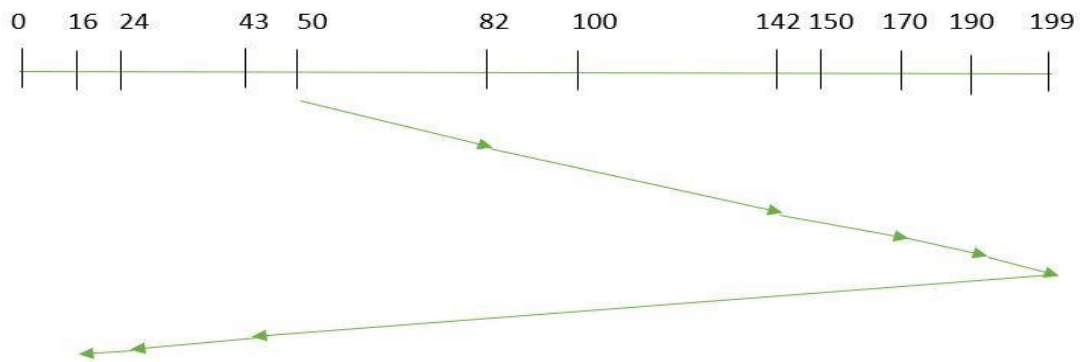


**Total OverHead Movement(Seek Time)=  $(50-43)+(43-24)+(24-16)+(82-16)+(140-82)+(170-140)+(190-170) = 208$**

### **Type3: SCAN Algorithm**

If the disk head is initially at 50 and if the head is moving towards forward direction, then it services 82, 142, 170, 190 till 199 data included.

-At cylinder 199 the arm will reverse and will move towards the other end of the disk servicing 43, 24 and then 16. The SCAN is also called as elevator algorithm.

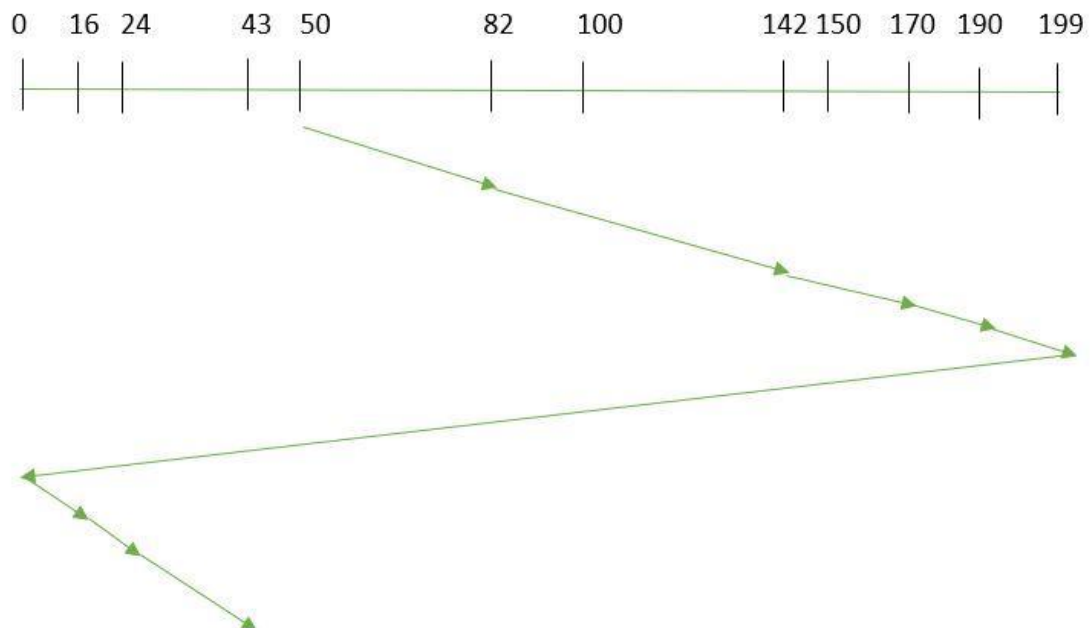


**Total OverHead Movement=  $(199-50) + (199-16) = 332$**

#### **Type4: C-SCAN Algorithm**

-If the disk head is initially at 50 and if the head is moving towards the outer track, it services 82, 142, 170, 190, and 199[reaches till end of the track].

-At cylinder 199 the arm will reverse immediately returns to the beginning of the disk without servicing any requests on the return trip (see chart below) and starts servicing again once reaches the beginning [including starting track zero] which serves 16,24 and then 43.



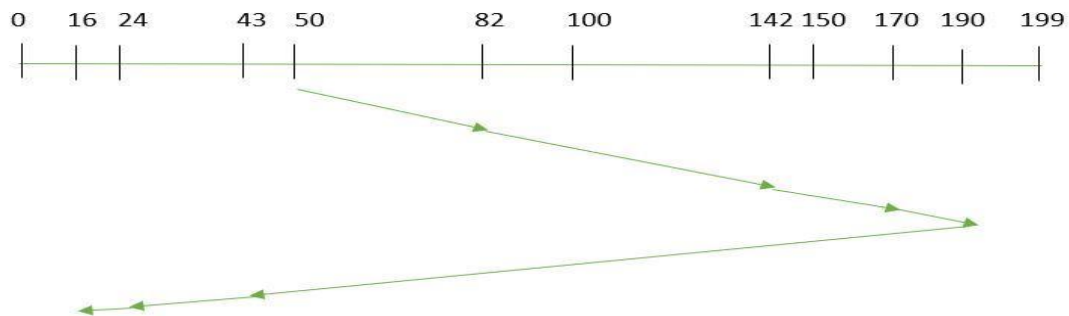
**Total OverHead Movement=  $(199-50) + (199-0) + (43-0) = 391$**

#### **Type5: LOOK Algorithm**



If the disk head is initially at 50 and if the head is moving towards the outer track, it services 82, 142, 170, 190[Here it does not reach 199 like SCAN].

-At the final request 190, the arm will reverse and will move towards the first request 43,24 and then serves 16[does not reach end track].



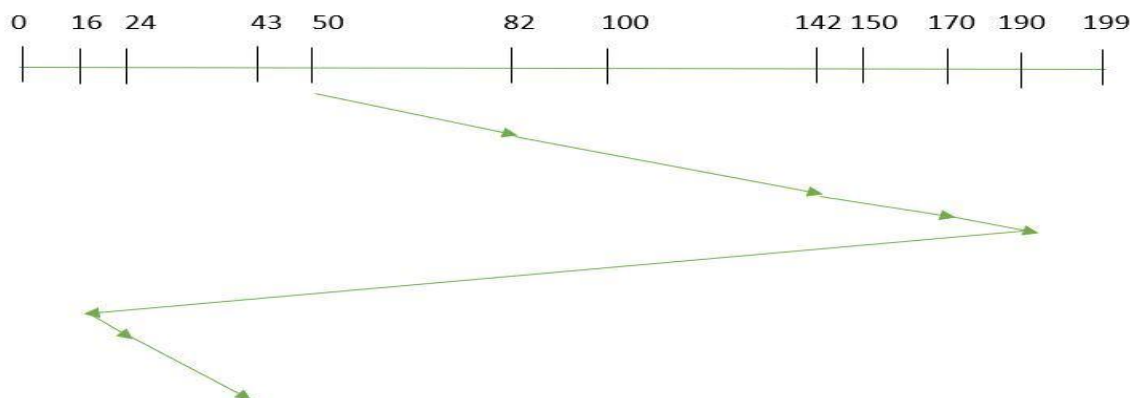
$$\text{Total Overhead Movement} = (190 - 50) + (190 - 16) = 314$$

#### Type6: C-LOOK Algorithm

Here the arm goes only as far as the final request in each direction. Then it reverses, without going all the way to the end of the disk.

-If the disk head is initially at 50 and if the head is moving towards the outer track, it services 82, 142, 170, 190 without reaching end tack 199.

-At the final request 190, the arm will reverse and will move towards the beginning of the first request 16 without starting from track zero and then serves 24, and then 43.



$$\text{Total Overhead Movement} = (190 - 50) + (190 - 16) + (43 - 16) = 341$$